

Unit Context:

This unit will take place in the Living Environment Regents Prep classroom with 13 students and 3 teachers. This class is fully in person and meets 4 times a week with a 60 minute class. These students have all previously taken and passed their living environment class but did not pass the New York State Regents Exam which is required for graduation. I plan on implementing this unit in class after students complete the January regents exam as a reminder of the importance of science in our lives regardless of the outcomes we may see on an exam.

All of the students in this class come from historically underrepresented populations. The school district where this unit is taking place has a diverse population and according to the New York State Education Department, the district's students are majority white with 18% Hispanic and Latino, 17% Black, 7% Multiracial and 5% Asian or Pacific Islander (NYSED,2023). 61% of students are economically disadvantaged which can change the way students view science as often technology (which is grouped with science) can be expensive and not affordable for these students (NYSED,2023). With this in mind when approaching a scientific topic it is important to create a connection back to something all students experience such as food from the cafeteria (which is free for all students at this school), riding the bus to school, snow, etc.

In our classroom we have built on students' cultural understandings of science by connecting scientific topics back to things students have told us they see in their lives such as their families, jobs, hobbies and other cultural activities. The goal of this mini unit is to allow students to connect science back to something they all see in their lives such as cooking. Regardless of if their families cook at home or if their only experience with cooking is what they see in the lunchroom students will have a point of connection with the topic. Many of these students struggle with reading but have expressed enjoyment when books are read to them. We have included reading skills in our class for the sake of helping students read and comprehend the questions being asked of them on the regents exam so certain skills like using context clues, finding connections to words and spelling out what we see have all been taught in this class.

Disciplinary Focus:

This lesson is building on previous science lessons about human impacts on the environment. Students have learned about ecology, impacts of land usage and fossil fuels. This lesson builds on those ideas while incorporating renewable energy sources

that we have yet to discuss. The goal is by the end of this mini unit students will have a stronger understanding of positive things humans can do to help the environment instead of just negative impacts caused by humans.

Essential Question: How does human energy usage impact the environment?

Justification: This unit will take place after students complete a high stakes exam so this unit is designed to focus on the bright sides of science. We have previously discussed how humans negatively impact the environment but this unit will shift to focus to what we can do better. All students have experienced cooking in some capacity even if it is just the dining hall during lunch hour at the school. Focusing on cooking will help my students learn through an entry point that is understandable for them instead of abstract science topics that can be hard to visualize. This will help to “demystify” the science and literacy learning that is occurring during this unit. For students who come from historically marginalized communities this different approach to learning will allow students to highlight their individuality, criticality and inspire joy.

Lesson 1: Impacts of science in our community

Objectives: Students will present an invention that had an impact on a part of the student's life

Activity 1 10-15 Minutes	Students will read “Ikbar and his Ingenious Invention” by Elizabeth Suneby. Ideally the school or community library would provide enough books for students to be able to read this individually if they so choose. In addition a youtube video of the book being read aloud will be provided to those who prefer multimodal ways of processing stories. This will also serve as a backup plan should physical copies not be possible.	Resources: Video Read Aloud of “Ikbar and his Ingenious Invention” by Elizabeth Suneby.  Iqbal and His Ing...	Focus on Joy- High school students often struggle with balancing wanting to be treated as adults while still enjoying being a kid. Asking students of this age to read a picture book can reignite their love for literacy and science through both words and pictures that allow them to feel like a kid while still being challenged to thinking critically. Focus of Identity- The book chosen highlights a character from a marginalized population that some of my students will connect to. This book inspires the readers to look at science solutions for their life similar to what Ikbar did.
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Activity 2 30 Minutes	Students will be tasked with researching a science or engineering invention that made a significant impact on their daily lives. This could be something they see in their community, their culture, a hobby or sport, or just something they are interested in.	Literacy Mini Unit... Teacher Created Example Teacher Example...	Focus on Criticality- Students are being asked to evaluate their community looking for examples of good or bad uses of science and engineering then evaluate that role that power played in this invention. Focus on Identity- Students will pick a part of their identity that they want to investigate inventions within. This could be from their community, culture, a hobby they enjoy, etc. Focus on Skills- Students will practice researching skills such as using trusted sources and citing references.
Activity 3 Rest of Class Time	All student's invention worksheets will be taped up around the room. Students will then use the rest of class time for a gallery walk where they look at each other's work and use post-it notes to comment on things they find interesting, something they learned or a question they have.		Focus on Intellect- Students will reflect on the work of their peers and comment/critique their posters. Focus on Joy- Students will interact with their peers and move around the room as they wish. We will have music playing to create a fun learning environment.
Assessment: The assessment for this lesson will be the student's invention worksheets. This is an informal formative assessment that will not be graded but will help me evaluate if they thought deeply about the impacts this invention and science has on their lives. They will use information learned since the beginning of this class to help guide ideas about inventions.			

Lesson 2: Energy Sources

Objectives: Students will research an energy source and the impacts that this energy source has on the environment.

Activity 1 20 Minutes	Students will split into groups to study energy sources. Each group pick an energy source then use the materials provided to help them learn about the energy source that they have been assigned. These materials will provide the scientific understanding of how that source works as well as the sociocultural implication of each energy source. Groups will jigsaw their materials. They will have 10 minutes to read their short articles then 10 minutes to share what they learned with their group. Energy sources studies will include - Solar - Geothermal - Natural Gas - Coal - Wind - Propane	 Literacy Unit Day... Readings are in the document linked above. Students will have access to these digitally and printed.	Focus on Joy- Students will pick which groups they want to work with and which energy source they find most interesting to learn about. Focus on Skills- Students will read the material on their energy source and summarize the most important pieces of information for their group mates. Focus on Criticality- Students will assess the materials used to determine what perspectives and voices are missing from the conversation. To this they are addressing unequal representation and power dynamics when it comes to the energy industry.
Activity 2 20 Minutes	Students will have 5 minutes to discuss what perspectives they think are missing from the articles they read. This could be blogs from those who live close to these power sources, workers, politicians, etc. Students will then have 15 minutes to find another source to add to their collection. This source could be an article, podcast, youtube video, etc.		Focus on Intellect- Students will use their intellect and previous knowledge to search for sources that provide a deeper story of the use of their energy source. Focus on Criticality- Students will evaluate the biases of the sources they want to use.

Activity 3 Rest of Class Time	Students will create a poster that shows what they learned about their energy source. This poster can be made using poster paper or online using Canva. The posters should include what the students feel to be the most important point of their energy source		Focus on Identity- This activity asks students to create a visual using any means that work best for the students. Students can represent how their energy source impacts their lives and what they learned from the articles. Focus on Joy- This activity provides a creative outlet for students that spark joy and imagination
Assessment: The posters created by groups will serve as the informal formative assessment for this activity.			

Lesson 3: Energy Debate

Objective: Students will make a claim about an energy source and use evidence gathered to defend their points and counter other's points.

Activity 1 10 minutes	Students will have the first 10 minutes of class to set up the posters that they had made during the previous class. They will then walk around the room and take notes on the information present in other students' posters. This information will be used as their evidence during the debate.		Focus on Skills: Students will create claims using evidence provided on the posters that they see around the room. Focus on Joy: Students will experience the creativity of their peers as they view each other's posters and learn together.
Activity 2 Rest of Class Time	Students will debate which energy source is the best. This could be for environmental reasons, justice, or efficiency. Each student will be given 2 pennies before the debate begins. Once students speak they will need to turn in a penny. After the whole class has used their pennies we will start the process over again. This is done in an effort to ensure all students participate equitably. If students do not want to speak verbally they can contribute to the padlet that will be on the board during the debate.		Focus on Intellect: Debates allow students to critique and respond to their peers by building on different perspectives that they all have. This critique and debate builds on students' intellect. Focus on Skills: Students will build skills such as making claims and providing evidence, persuasive literacy and public speaking.

Assessment- After the debate students will fill out an exit ticket that will serve as a summative assessment based on the lessons they've learned this week. This assessment gauges if students are able to make a claim and use what they have learned to back up that claim. The exit ticket and a teacher example are included below.

 Energy Debate Exit Ticket

Lesson 4: Science in Action Objective: Students will build a solar oven using scientific and engineering design principles.			
Activity 1 45 Minutes	Students will have the entire class time to build their own solar oven out of recyclable materials. An example with step by step instructions will be provided to students who may feel overwhelmed with the idea of building something from scratch.	Example Solar Oven  Solar Oven Exam...	Focus on Intellect: The “Power of Doing” (Muhammad, 2020, P.111) focuses on intellect by allowing students to put their learning into action. Focus on Skills: Students will use their engineering and science skills that they have been building over the course of the class such as designing, trial and error, etc.
Activity 2 15 Minutes	Students will create a crayon mold out of tinfoil. This mold can be any shape such as letters, heart, etc. Students will then pick old crayons and arrange them in their molds in any pattern. Once the crayons melt in the solar ovens (Takes 2-3 days) students will have a personalized crayon.		Focus on Identity- Students can make a crayon that expresses who they are whether that be in the mold of the crayon or in the colors used. Focus on Joy- Creating something allows students to express themselves, have fun and a take home souvenir to show their families.
Assessment: Student’s working solar oven will serve as a formative assessment to see if students understood the scientific principles behind what makes a solar oven cook.			

Personal Reflection:

Something we frequently discuss in science teacher prep courses is how to “demystify” science which is often thought of as a difficult and hard to understand field. I believe that step one to this process is to find a connection between the science and something students experience in their everyday lives. When talking about energy my first thought was to focus on cooking as food is essential for life. It should be a point of

connection between all of my students. This was the easiest part of designing this unit for me.

I wanted to start my unit with an interesting hook and decided a story would be a great starting point. We have been exploring reading with these students especially focusing on the possibility of students being able to have exams read to them. During this process students expressed liking experiences where they get to have things read to them as it allows them to comprehend the material. I was looking for a book that would highlight a nonwhite character that had a read aloud option as well as a book with images for those students who prefer visuals. This was probably the hardest part of the process especially because I knew I wanted to focus on cooking. In the future I would like to ask the school librarian to help me find books that are both representative of the population I work with and on topics for the science we are learning. During the planning of this unit it didn't occur to me to use other educators in the building as resources.

After finding the book that I planned to use as my hook my brain came up with a pretty natural progression based on what the character in the book learned. I knew I wanted to end my unit to have the students build a solar oven. I have done this previously with students but we cooked s'mores. I originally had the idea of students using their oven to cook food from their culture but this was tricky to implement. Melting crayons allows us to achieve a similar impact but provides students with something to take home which I love as it can spark conversation with their families. The crayon also creates another connection to identity.

I found identity was the hardest pursuit for me to incorporate into this science unit. I was originally most worried about incorporating criticality but this came more naturally than I expected. Criticality was able to build in by asking students to question their own communities, why certain inventions were or weren't successful and who those inventors were. The identity piece was a bit harder because while they had many opportunities for creative expression I wanted students to really show who they were through science. In the future I think starting with the end product in mind, connecting that end product to identity then working backwards to build the unit would be successful for me.

The pursuits of intellect and skill seemed to come naturally within researching and scientific processes though at first I did struggle with differentiating between intellect and skills. While I think these were included somewhat naturally I did need to dig a little deeper in order to develop these activities and ensure that the students were getting what I wanted them to from the activities. *Cultivating Genius* was a great resource to refer to for ideas on how to strengthen these pursuits. I have always believed that joy is one of the most important aspects of a classroom. Part of cultivating joy in a classroom for me starts with building a community within the classroom which can overtime spark joy in itself.

Overall I loved the challenges that came with developing this mini unit with all 5 pursuits in mind. In the future I would like to do this while also focusing on backwards design to help me start my planning with the outcomes I want students to produce. I also think I could have leaned on my peers and mentor teachers in my placement to get their advice on how to do this within the science field which is often viewed as rigid and difficult especially for students who have been placed in an extra class only because they failed a high stakes exam.

References:

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Muhammad, G. (2020). Cultivating Genius: An Equity Framework for Culturally and Historically Responsive Literacy. Scholastic